Introduction to HEC-RAS with a simple example

Information Required for Model

- GIS Model of the Terrain
- Boundary Conditions

Upstream:
Normal Flow
Critical Depth
Rating Curve (Q-h)
Flood Hydrograph
(Unsteady Flow)

Downstream:
Normal Flow
Critical Depth
Rating Curve (Q-h)

- Bed Specifications:
  - ✓ Bed Levels (cross section for 1D modelling, levels for 2D)
  - ✓Resistance numbers such as Manning n or Chezy ks

Example

A river (name Project River) is 1000 m in length (From downstream to upstream end).

A change in the gradient of the river occurs 400 m from the downstream end of the river. The bed levels at the downstream, 400 m from downstream, and upstream end is 1m, 3m and 9m respectively. The river has no bends or turns.

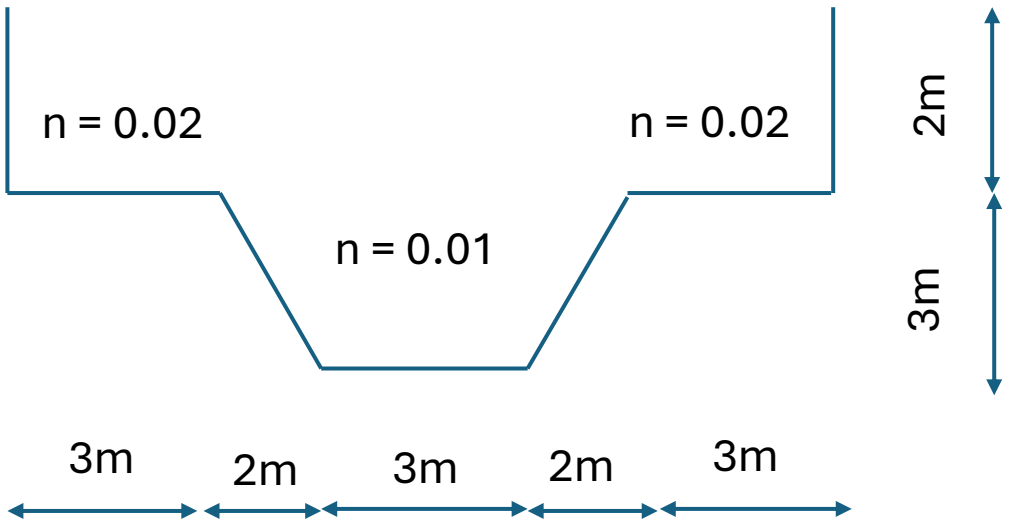
The main channel and flood plains has a Manning n value of 0.01 and 0.02 respectively.

Consider the following (Steady Flow Analysis):

Another river converges with Project River 400 m from the downstream end. The flow rate changes from 100m3/s to 150m3/s at the convergence.

Assume normal flow conditions at the upstream and downstream ends

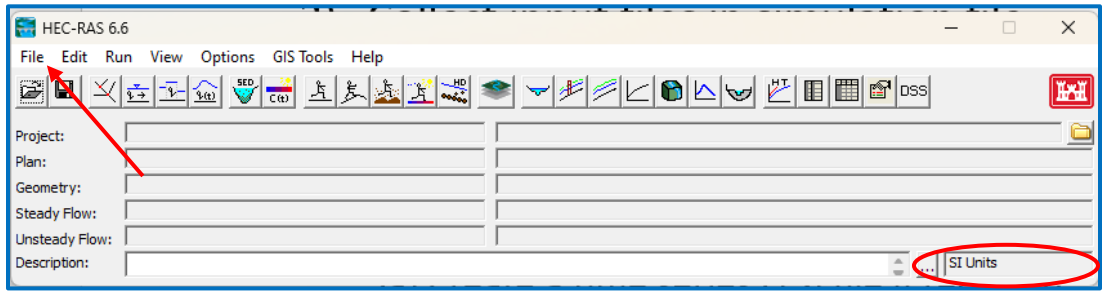
River Cross section (Remains constant)

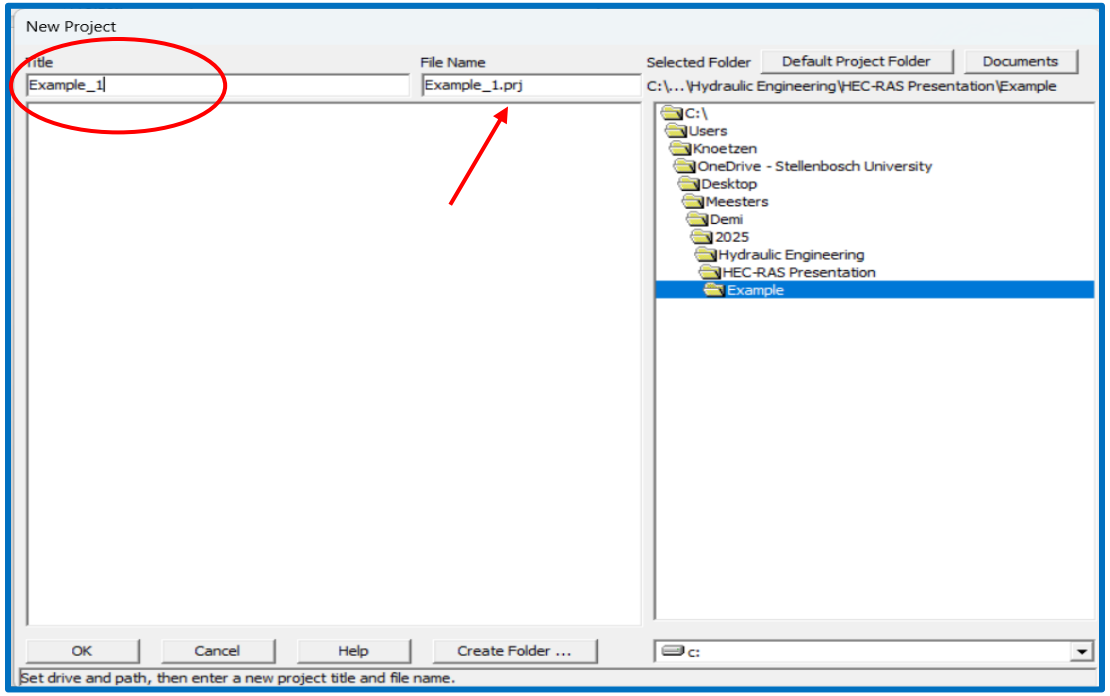


Outline of Steps:

1. Create a New Project
2. Add a River Reach
3. Add/Edit Geometric Data
4. Edit Steady/Unsteady Flow data
   - For Steady Flow:
     - Set Flow Rates at downstream and upstream chainages (stations)
     - Set downstream and upstream boundary conditions
5. Perform Steady Flow Analysis

1. Create a New Project

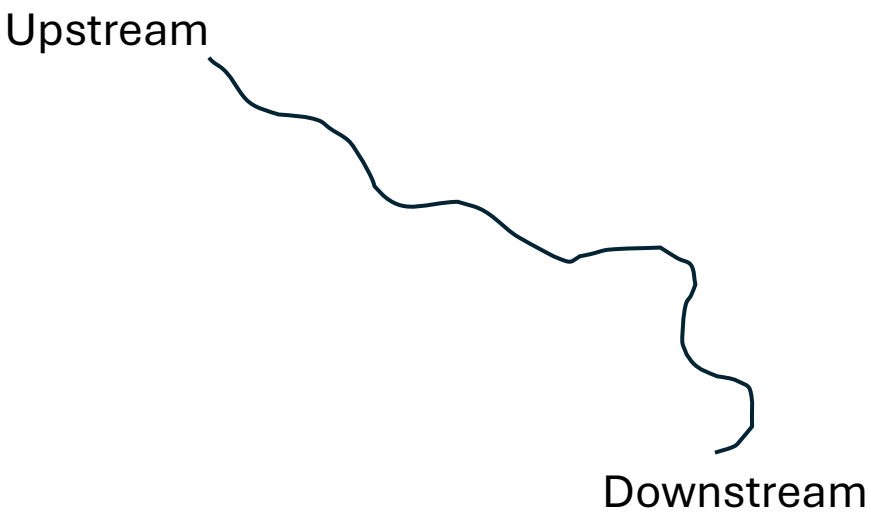




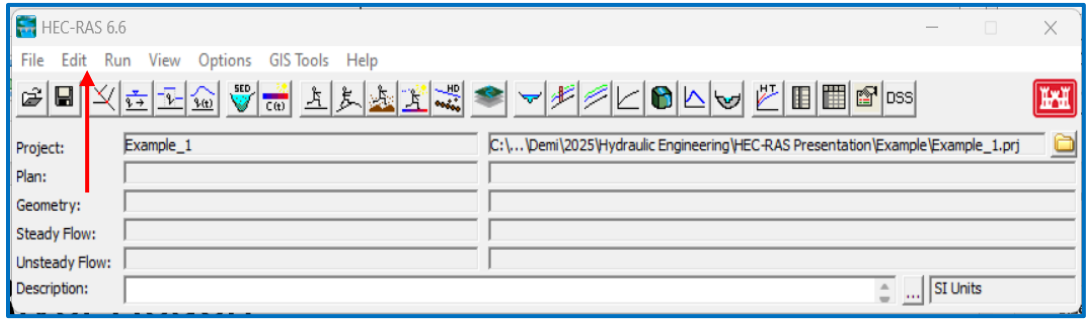
- Firstly check that the program operates in **SI Units**
- To create project:
  - **File -> New Project**
  - Insert the **title** of your project
  - Save the project into a folder easily assessable
  - Click **OK**
  - The project will be saved as a ***.prj** (In this case Example_1.prj)

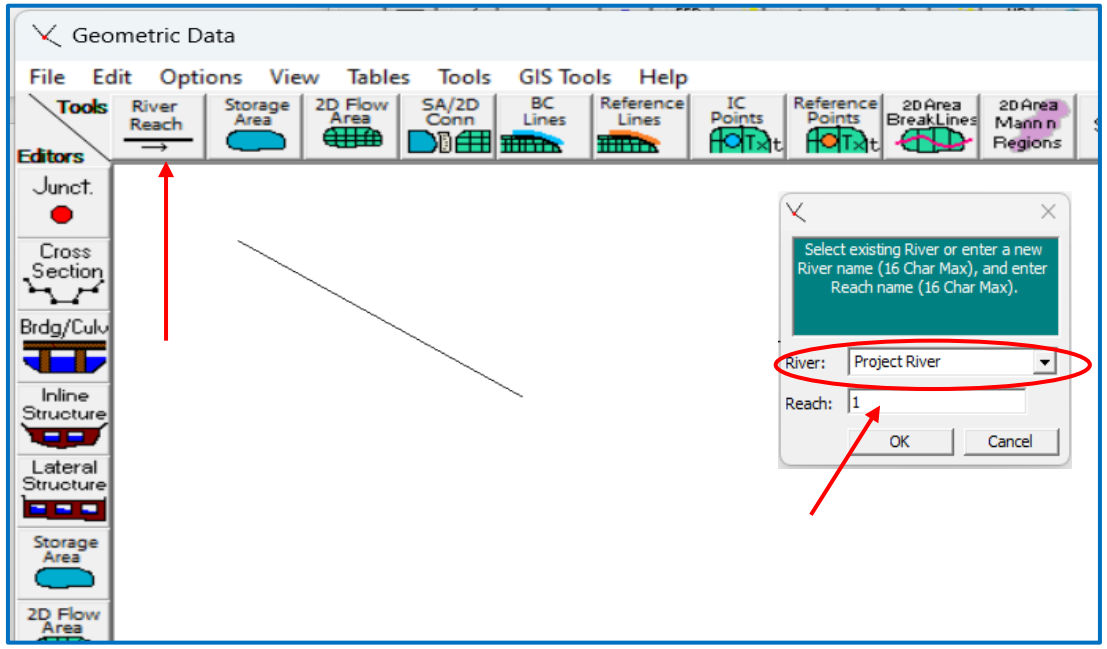
2. Add a River Reach

- A river reach is a continuous section of a river that can be modelled as a unit
- Added in Two ways:
  - ✓From Scratch
  - ✓From RAS-Mapper



Add a River Reach

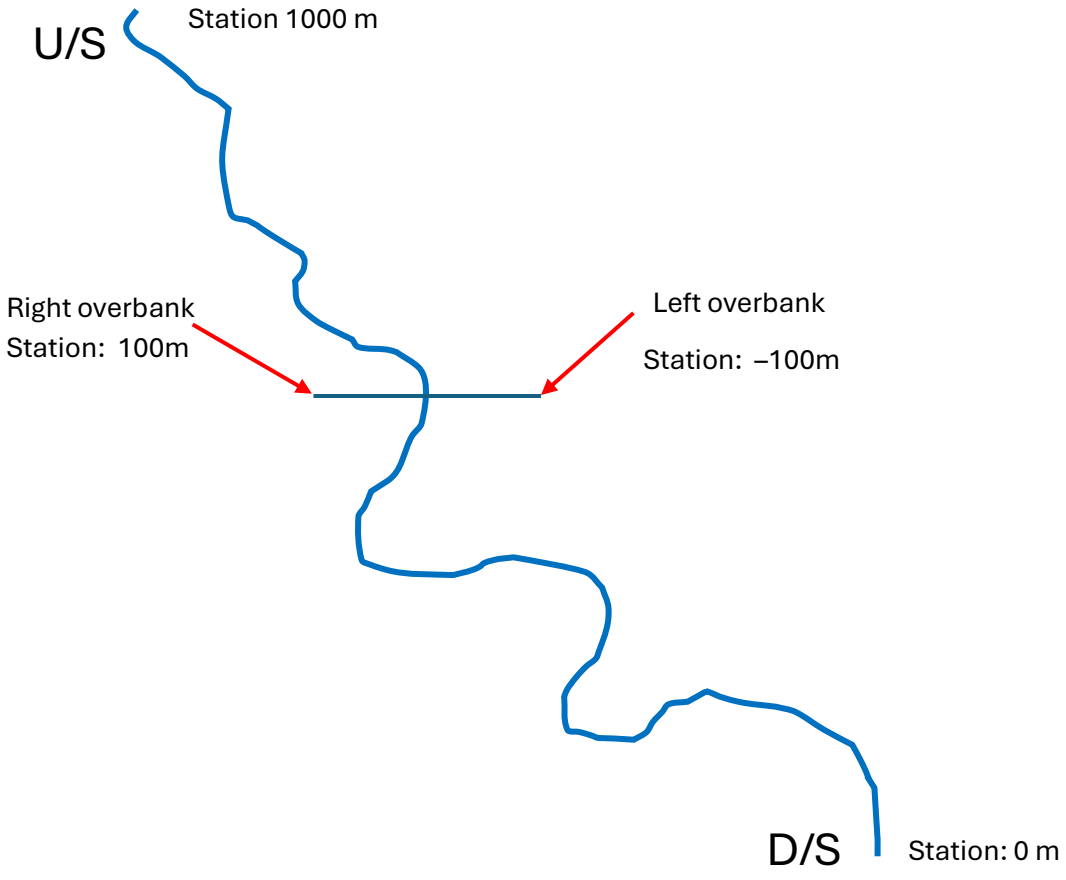




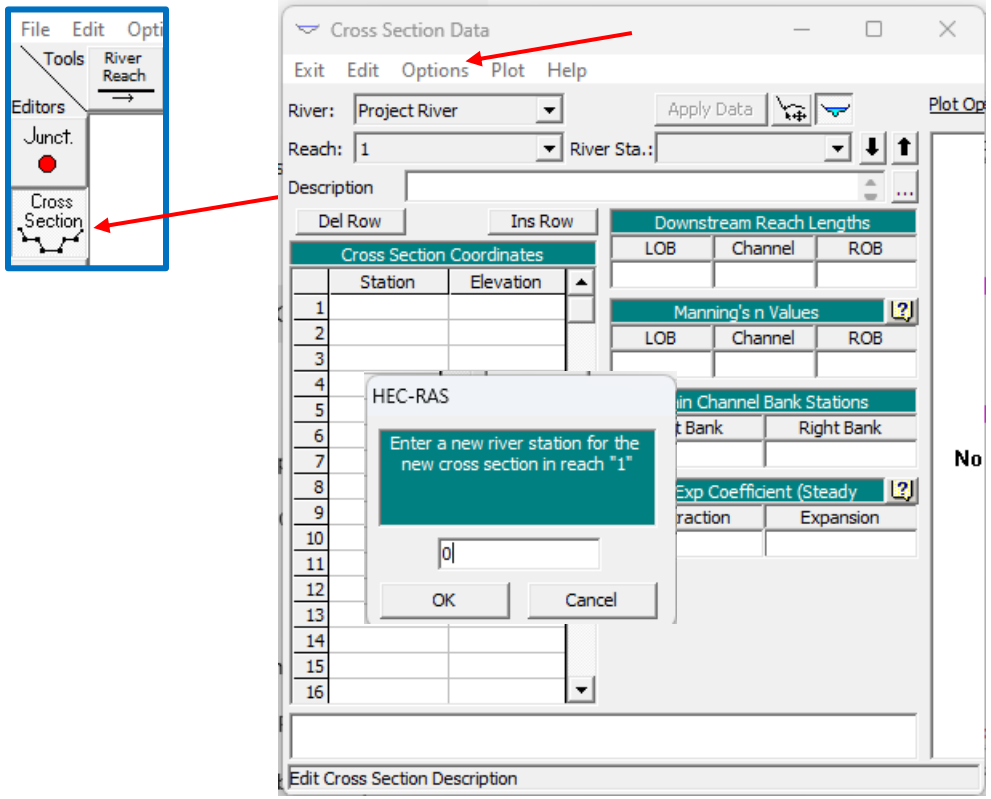
- Go to **Edit** -> **Geometric Data**
- Then click on **Add new River Reach**
- From Scratch:
  - Draw your desired river
  - Double click to end river reach
- Define the River
  - Provide a name for the **River**
  - Set a **Reach** number (if you have only one river, set a reach value of 1)

Add/Edit Geometric Data

- Introduction of the concept "Station"
  - Refers to the lateral distance used to define the location of points along a cross-section
  - Along the length of the river, it measures from downstream to upstream
  - And measures from left to right along the cross section of the river (left overbank – right overbank in the direction of flow)

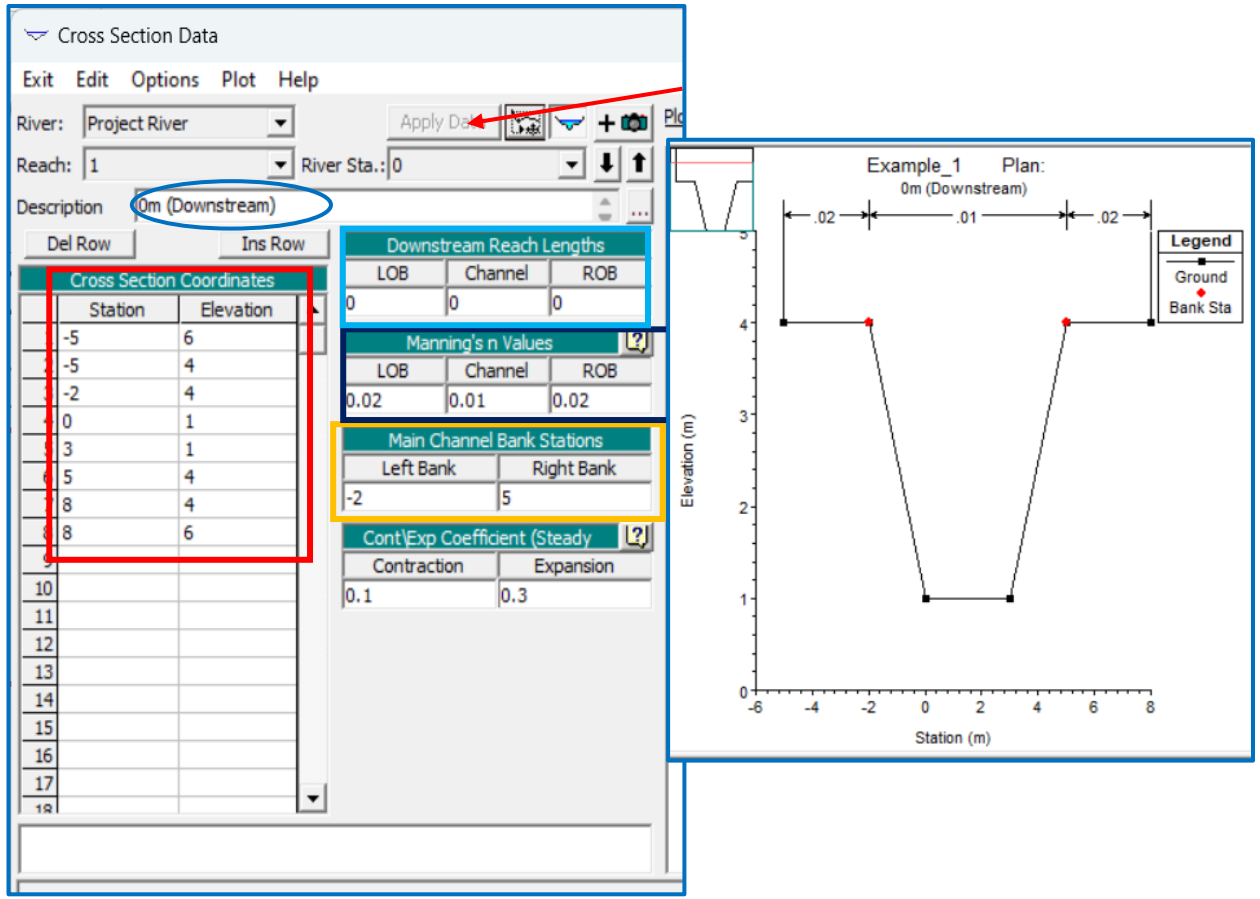


Add/Edit Geometric Data



- In Geometric Data, click on **Cross Section**
- Then click on **Options** -> **Add a new Cross Section**
- Enter the downstream station value (value of **0** m)

Add/Edit Geometric Data

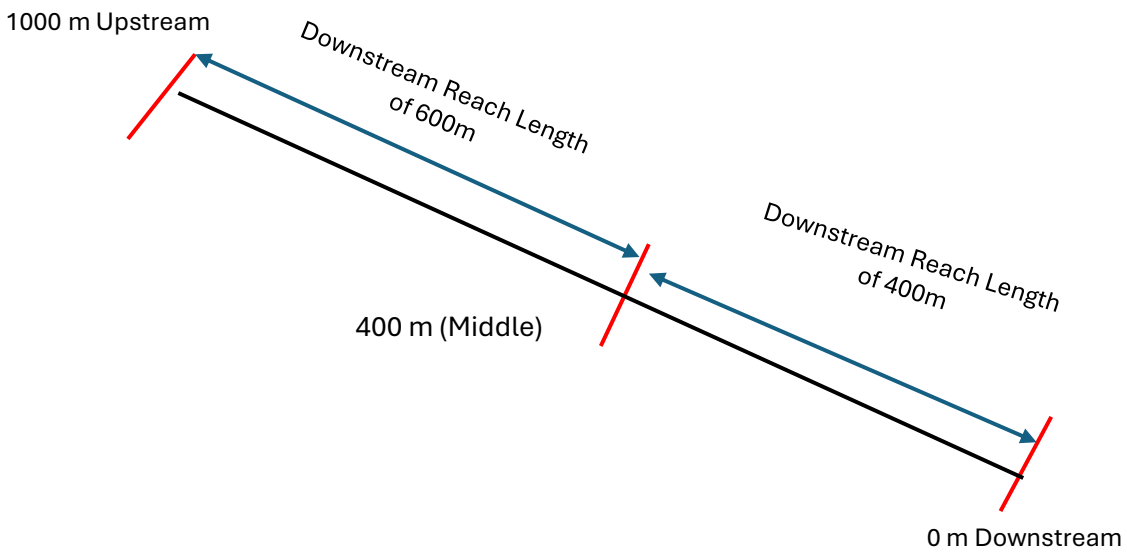


- Add river **description** (Make it logical)
- Add **Cross Section Coordinates**
- Identify the **left -** and **right bank** stations of the main channel
- Add main channel, left and right overbank **Manning's n values**
- Add **Downstream Reach Length** (at the downstream station, length is 0m)
- Now **Apply Data**

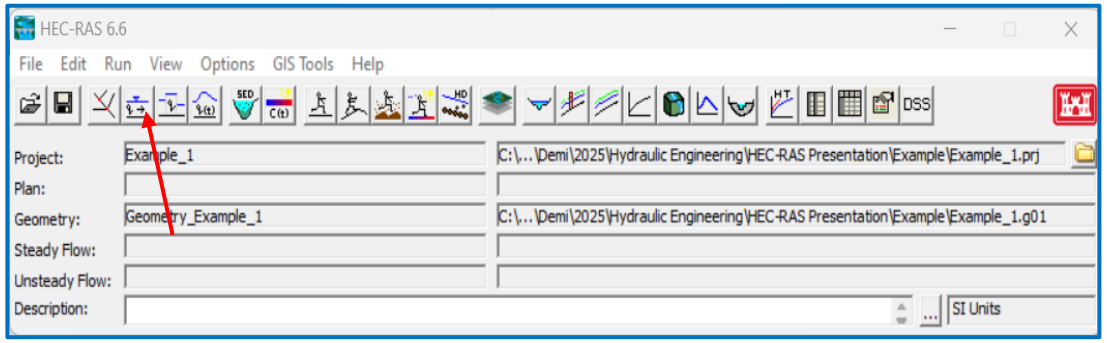
Add/Edit Geometric Data

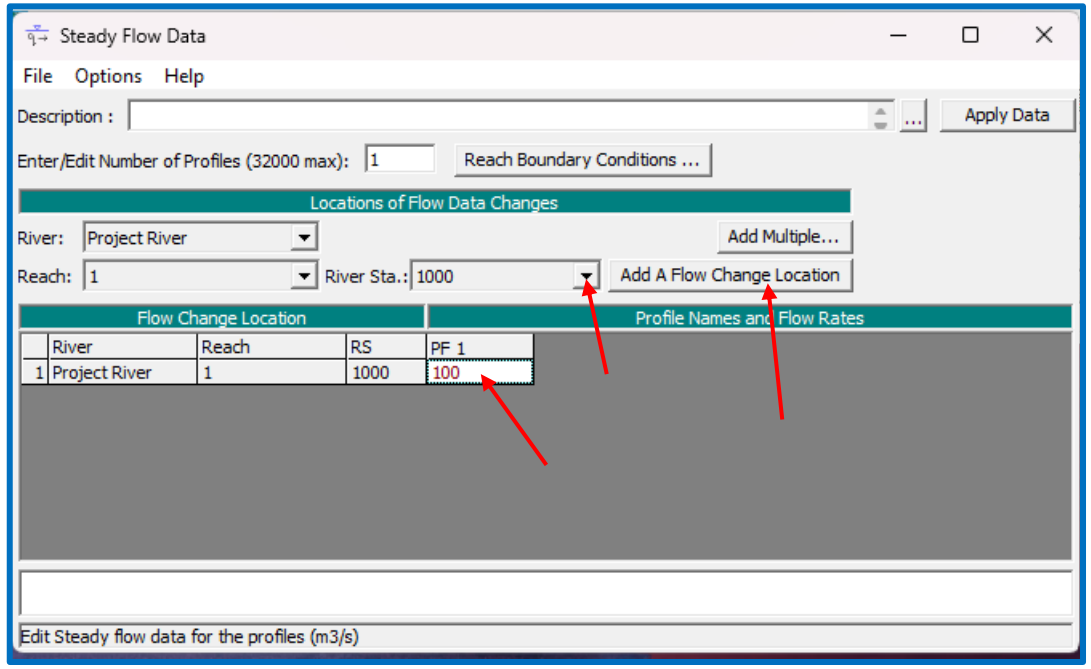
- For next cross section on the river reach you click again on Options -> **Add a new Cross Section** or one can **Copy Current Cross Section** and **Adjust Elevations (**after clicking on **Options** Tab**)**
- Important Concept regarding the Downstream Reach Lengths:
  - Represents the distance along the rivers' flow path from a given cross-section (river station) to the next downstream cross-section within the same river reach
  - When river has no bends LOB = Channel = ROB
- In this example: 3 cross sections station and downstream reach length reach values:

Description	Station (remember, refers to distance from downstream end of river)	Downstream Reach Length
0 m (Downstream)	0	0
400 m (Middle)	400	400
1000 m (Upstream	1000	600

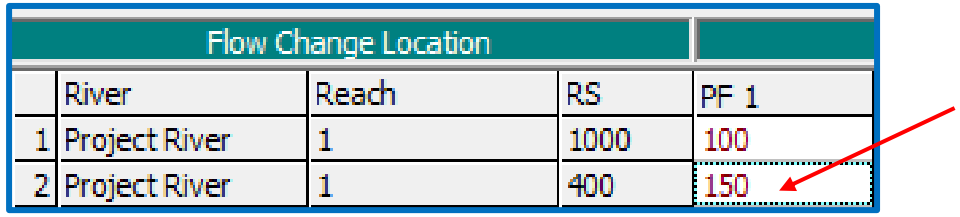


Add/Edit Steady Flow Data





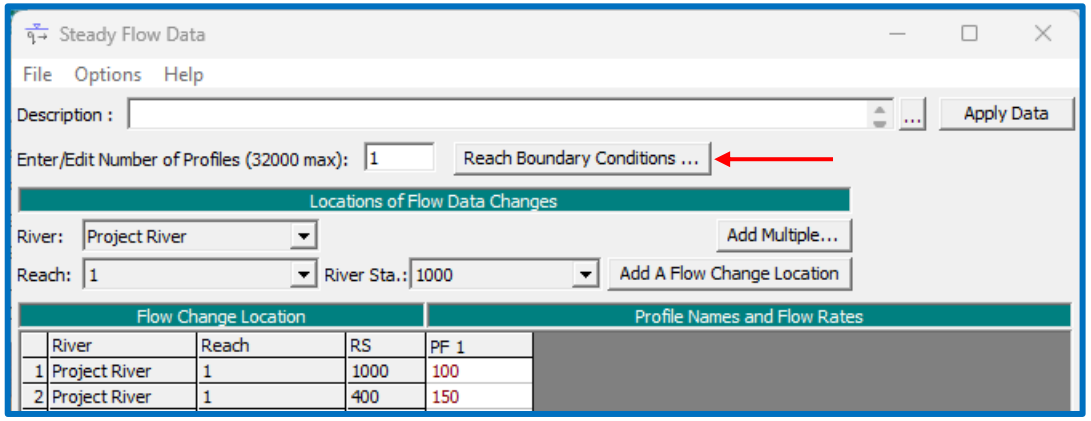
- Click on **Steady Flow Data**
- Firstly enter the upstream flow rate at **PF1** [in this case 100 m3/s at Station 1000m]
- If another river converges and the flow rate changes:
  - Go to the drop down button at **River Station** and select the station
  - click on **Add a Flow Change Location**
  - Then enter the flow rate the river reach changes to (Flow rate of river reach changes from 100 to 150m3/s at station 400 m

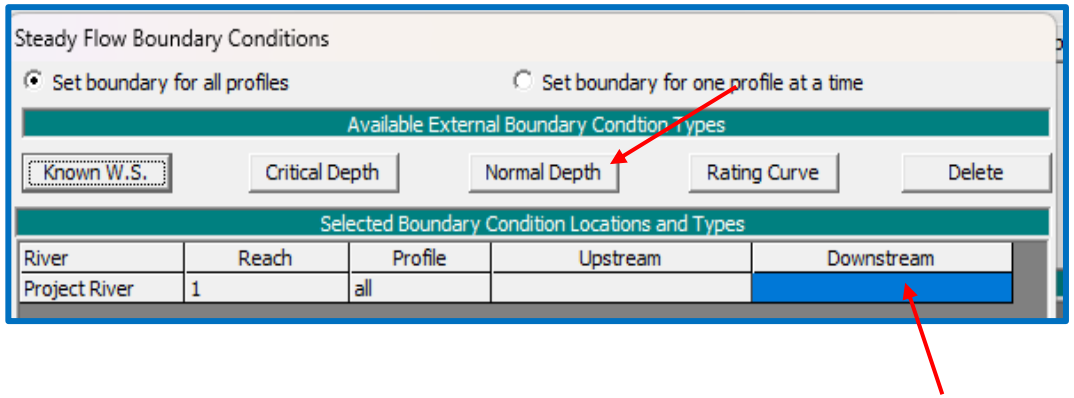


Flow Change Location

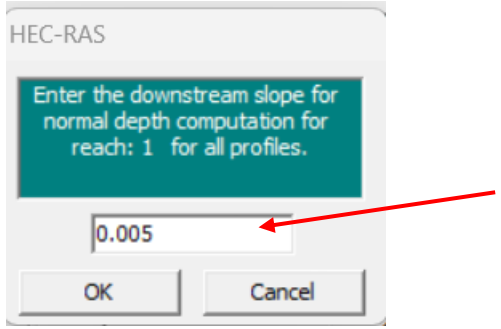
	River	Reach	RS	PF 1
1	Project River	1	1000	100
2	Project River	1	400	150

Add/Edit Steady Flow Data

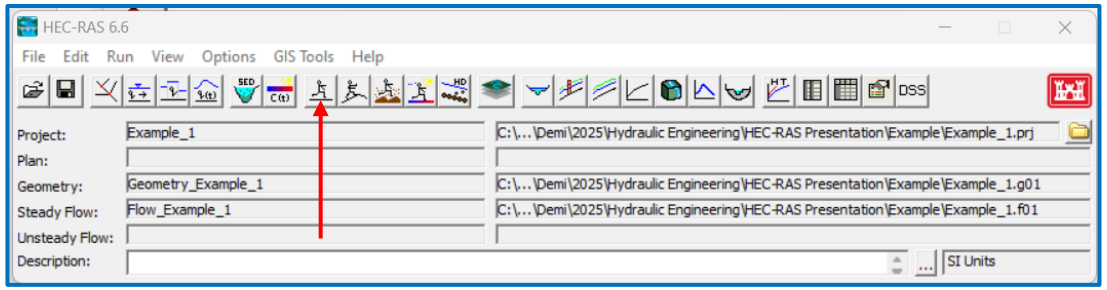


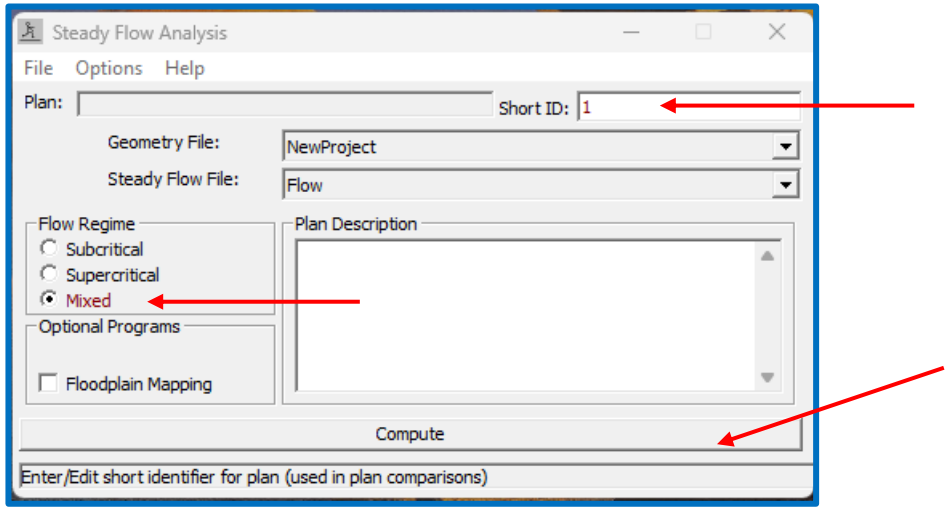


- Next click on **Reach Boundary Conditions**
- NB **Downstream** and **Upstream** must have boundary conditions
- If one assumes **Normal Depth** at downstream boundary conditions:
  - Add the slope at the downstream end of the river



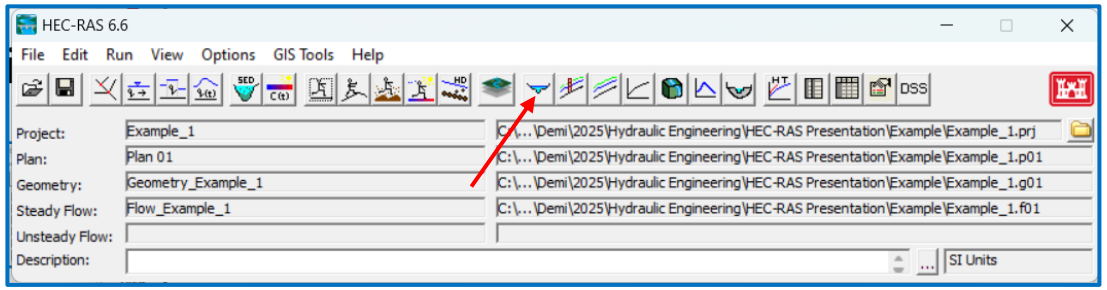
Perform Steady Flow Analysis

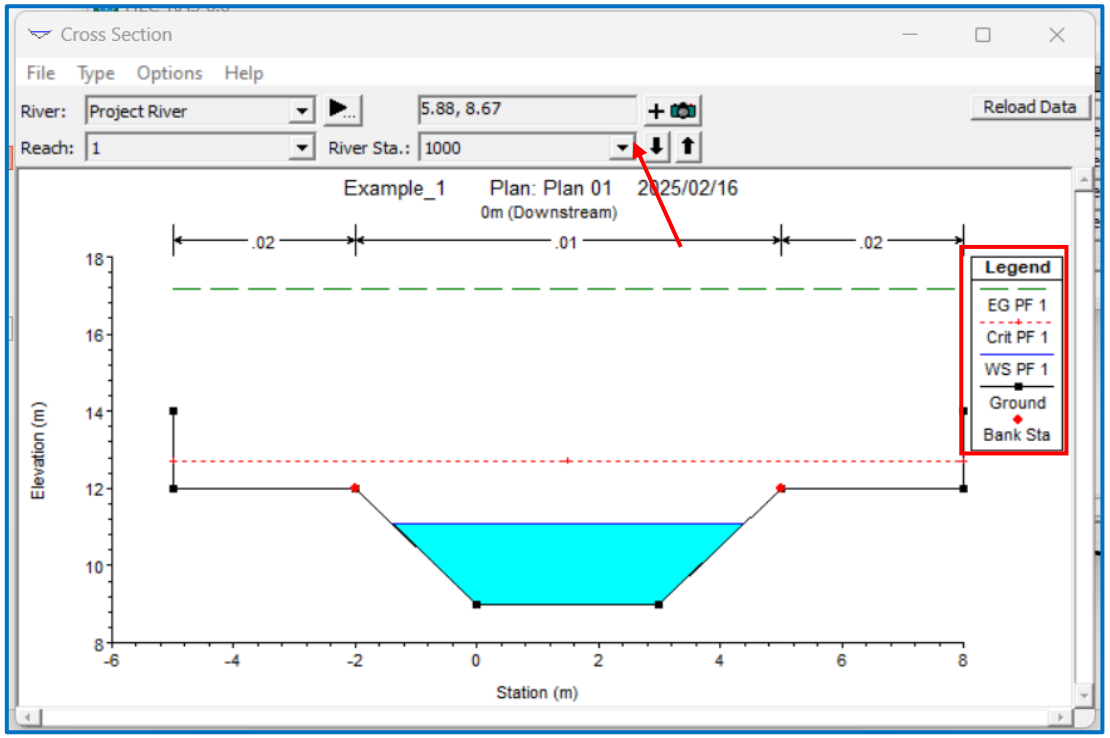




- To start the analyses click on **Steady Flow Analysis**
- Ensure flow regime is **Mixed**
- Enter a **Short ID**
- Then Click **Compute**
- This is where errors can appear, usually mistakes were made at Geometric Data and Steady Flow Analysis

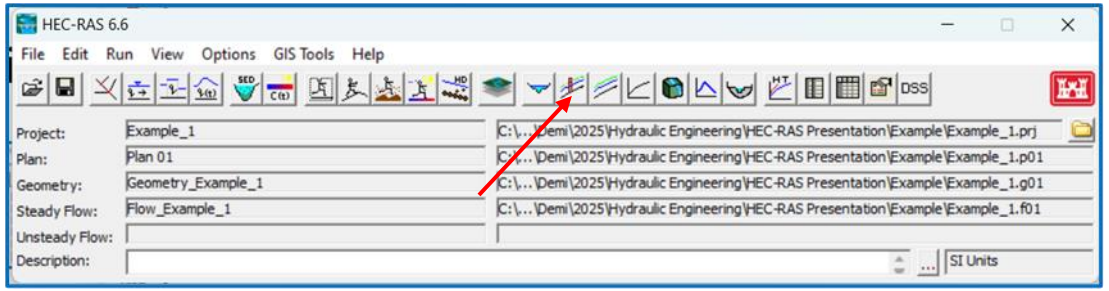
View Results of Steady Flow Analysis



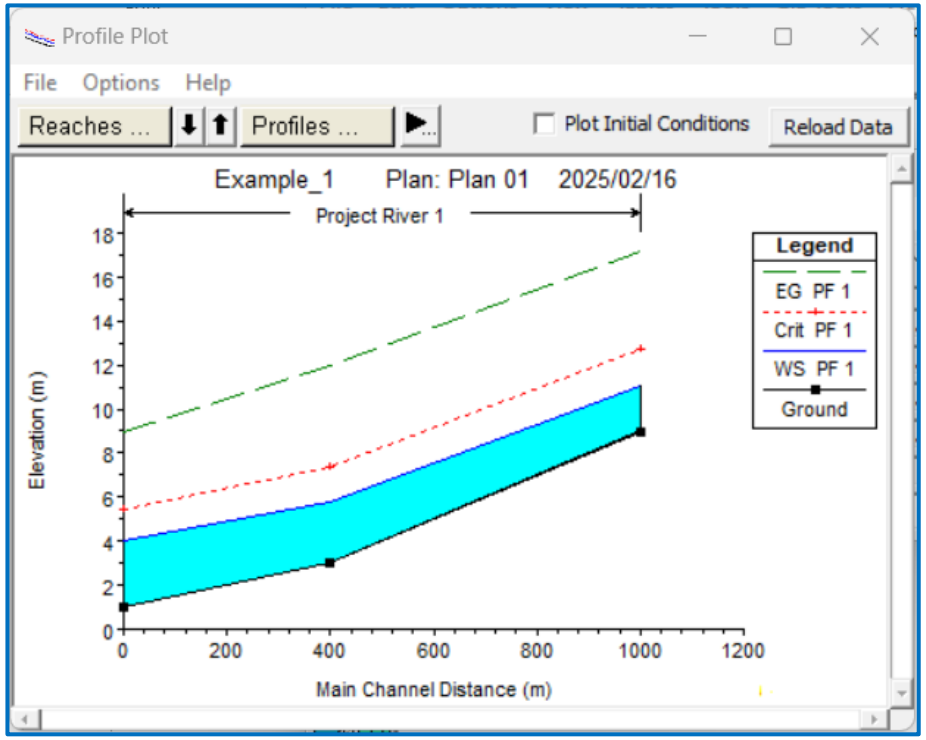


- To view the water depths at each the cross-sections, click on **Cross Sections**
- The cross section views water level, critical depth and energy level
- To view the water depth at another station, click on drop down button at **River Station**

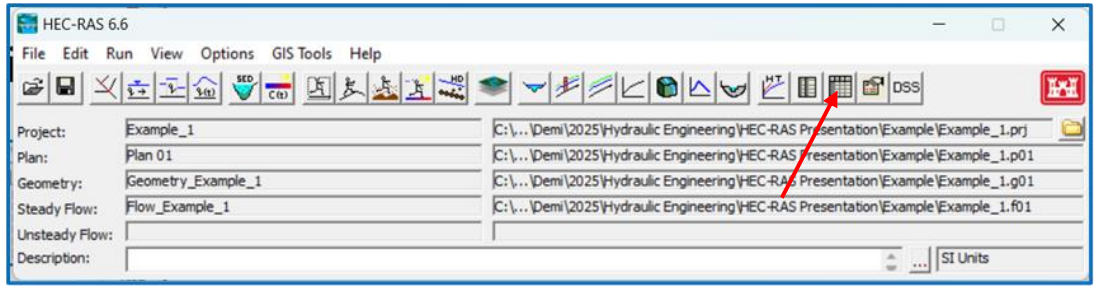
View Results of Steady Flow Analysis

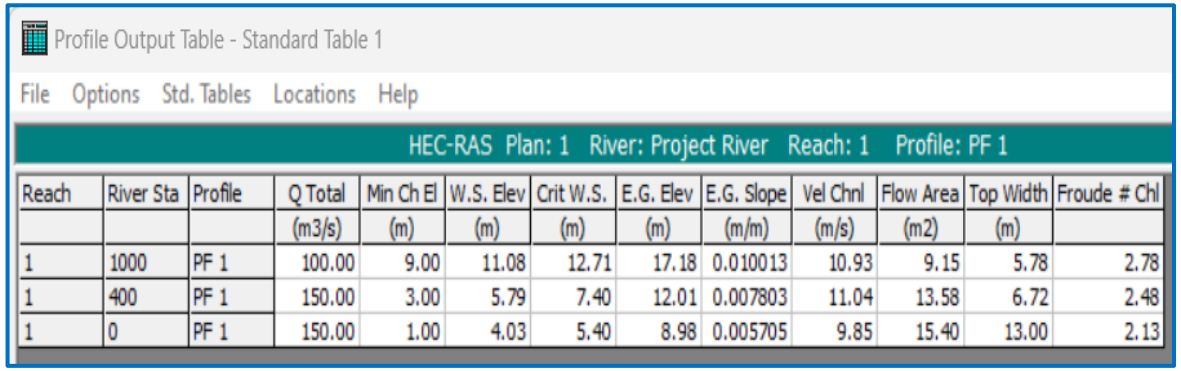


- To view longitudinal section profile click on **Profile Plot**



View Results





Profile Output Table - Standard Table 1

File Options Std. Tables Locations Help

HEC-RAS Plan: 1 River: Project River Reach: 1 Profile: PF 1

Reach	River Sta	Profile	Q Total	Min Ch El	W.S. Elev	Crit W.S.	E.G. Elev	E.G. Slope	Vel Chnl	Flow Area	Top Width	Froude # Chl
			(m3/s)	(m)	(m)	(m)	(m)	(m/m)	(m/s)	(m2)	(m)	
1	1000	PF 1	100.00	9.00	11.08	12.71	17.18	0.010013	10.93	9.15	5.78	2.78
1	400	PF 1	150.00	3.00	5.79	7.40	12.01	0.007803	11.04	13.58	6.72	2.48
1	0	PF 1	150.00	1.00	4.03	5.40	8.98	0.005705	9.85	15.40	13.00	2.13

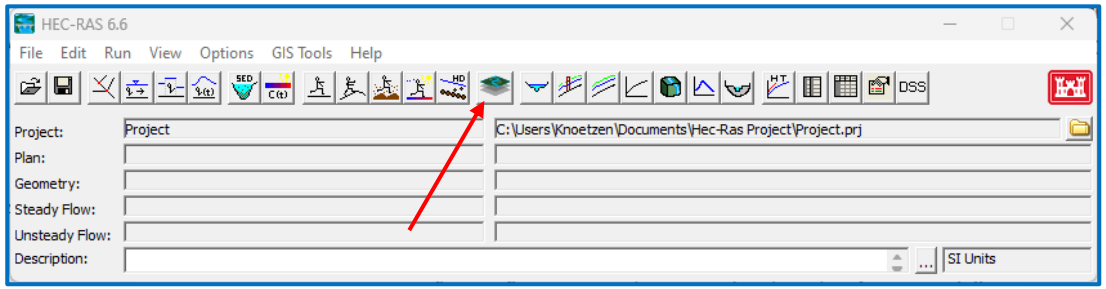
- To have a numerical output of the results click on **Profile Output Table**
- Shows the results of the water surface elevation, critical depth elevation, energy line elevation, channel velocity, etc.

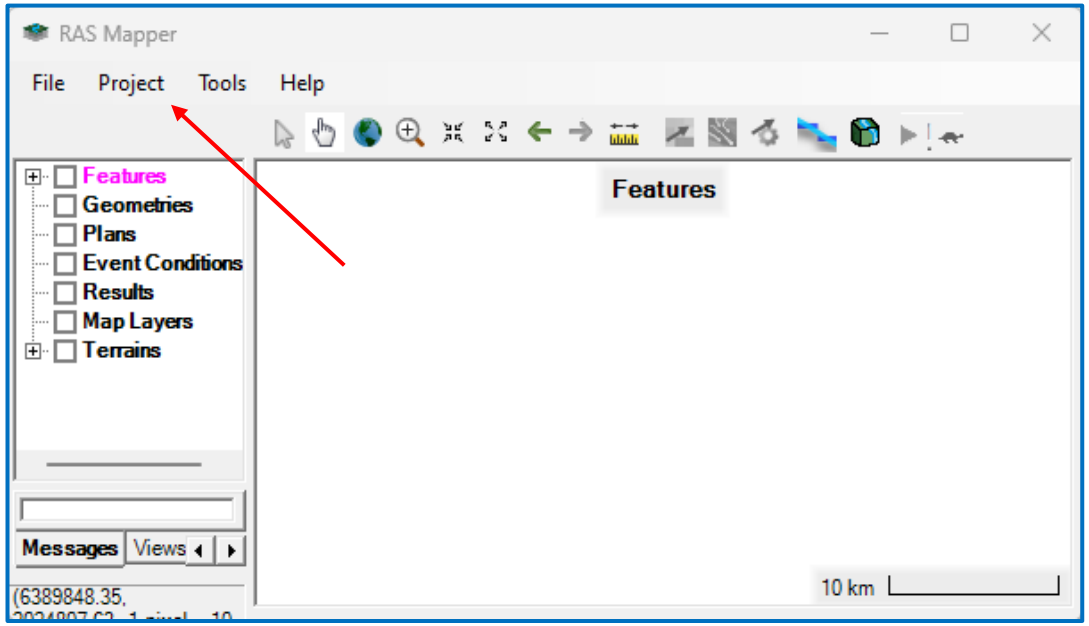
HEC-RAS Mapper

RAS Mapper

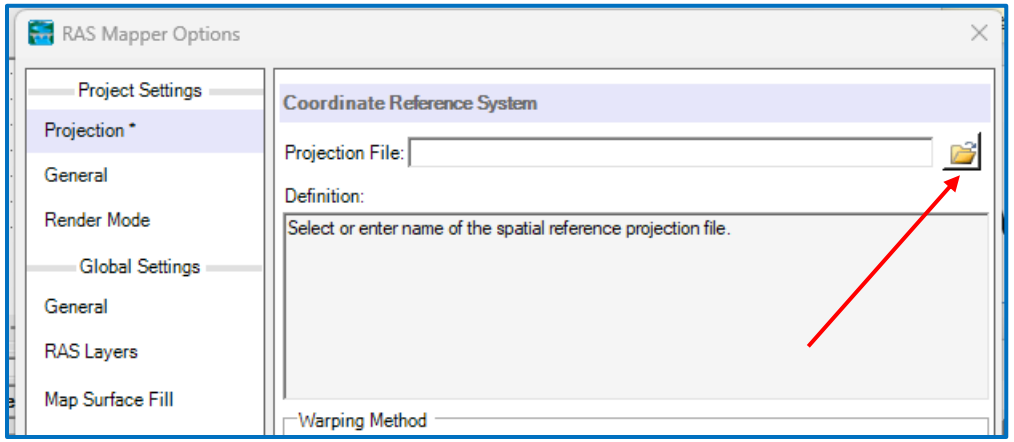
- RAS-Mapper is a GIS-based tool within HEC-RAS that allows users to create, edit, visualise and analyse spatial data for hydraulic modelling
- Files required:
  - Projection Files
  - Terrain Files (GeoTiff files)
- The above files are created using GIS software such as QGIS, ArcGIS, etc.

Add a River Reach: RAS Mapper

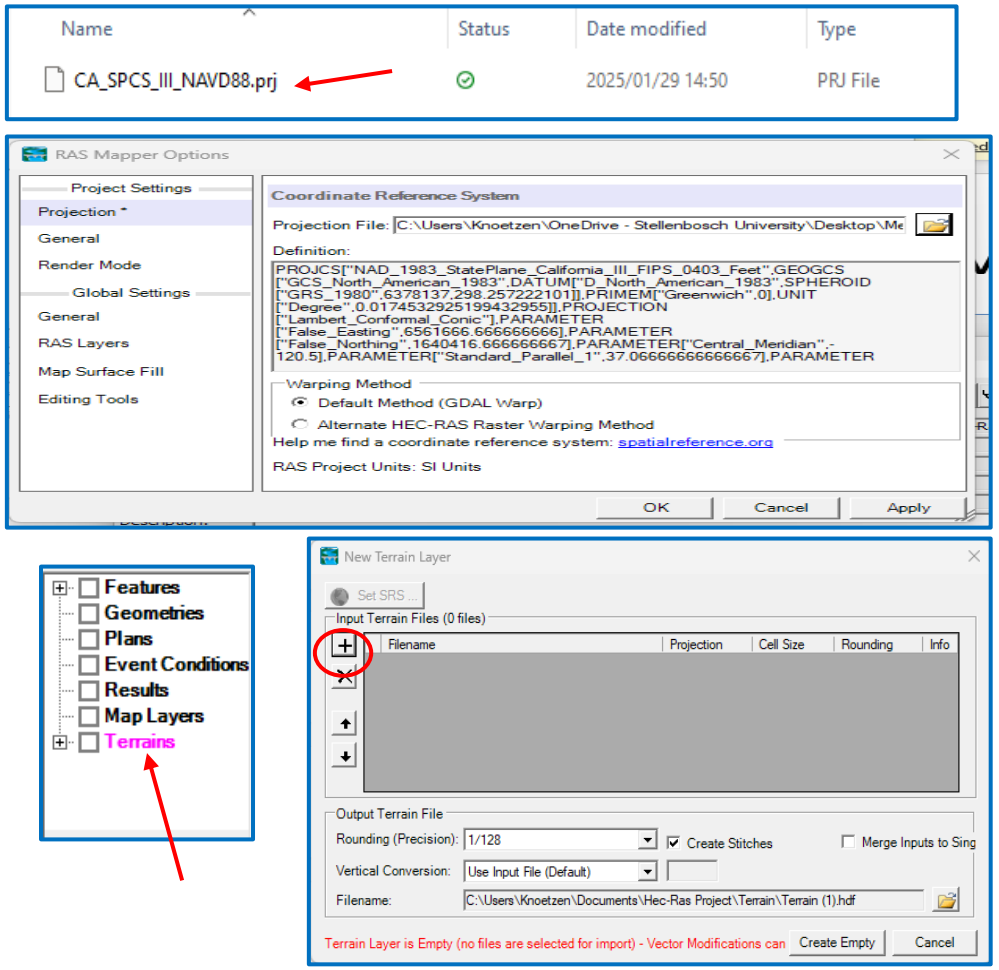




- Open **RAS Mapper**
- Click on **Project** -> **Set Projection**
- Import a projection file

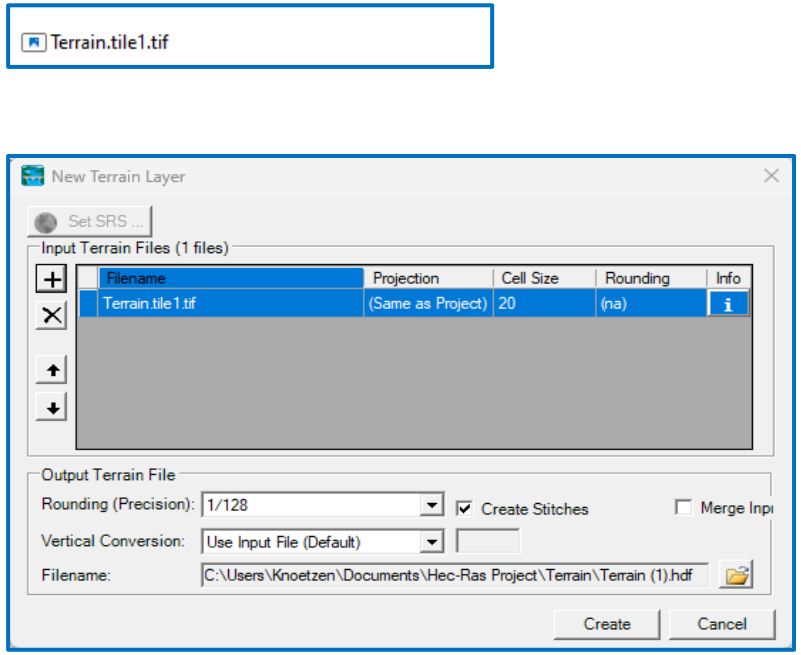


Add a River Reach: RAS Mapper

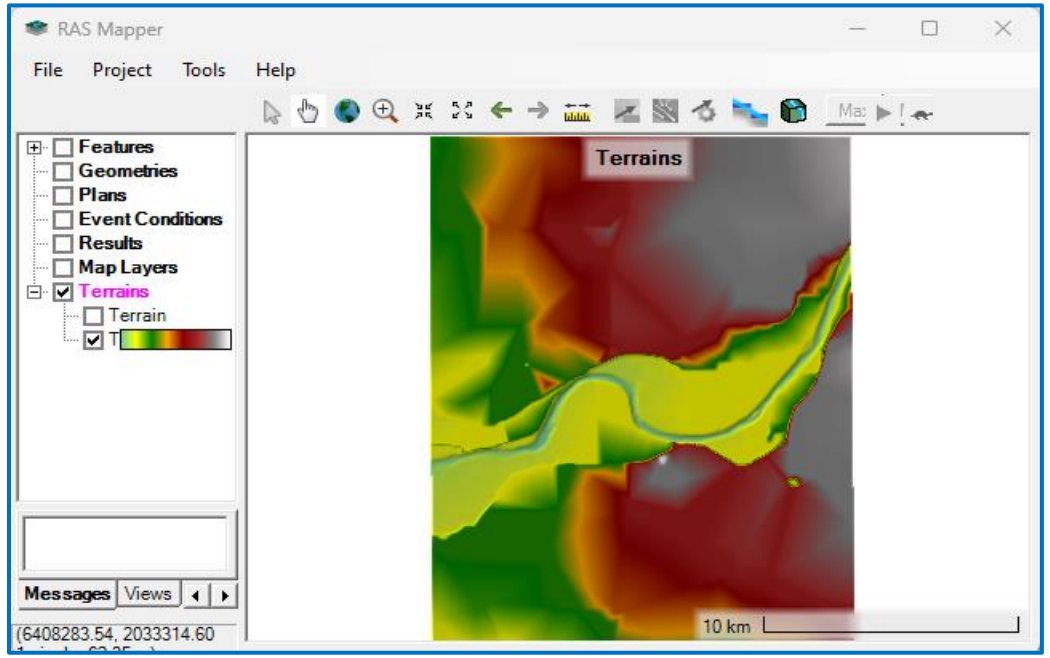


- Projection file is also in ***.prj** format (NB: Do not confuse with Project file)
- Click **OK**
- Next, right-click on **Terrains** -> **Create a New RAS Terrain**
- Click on "**+**" to import GeoTiff (***.tif**) file

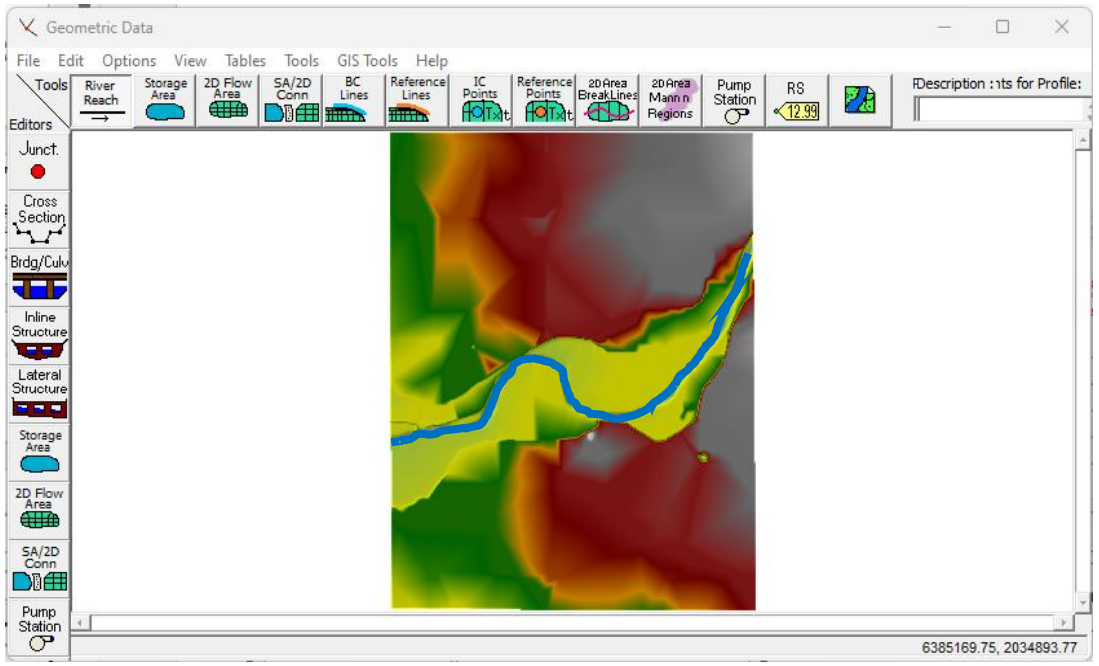
Add a River Reach: RAS Mapper



- Select ***.tif** file
- Click on **Create**
- GIS Map is created

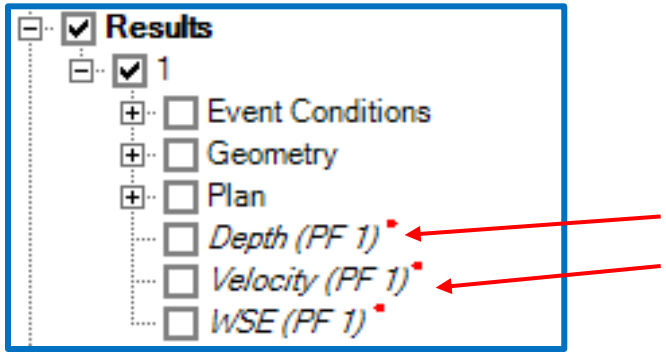


Add a River Reach: RAS Mapper



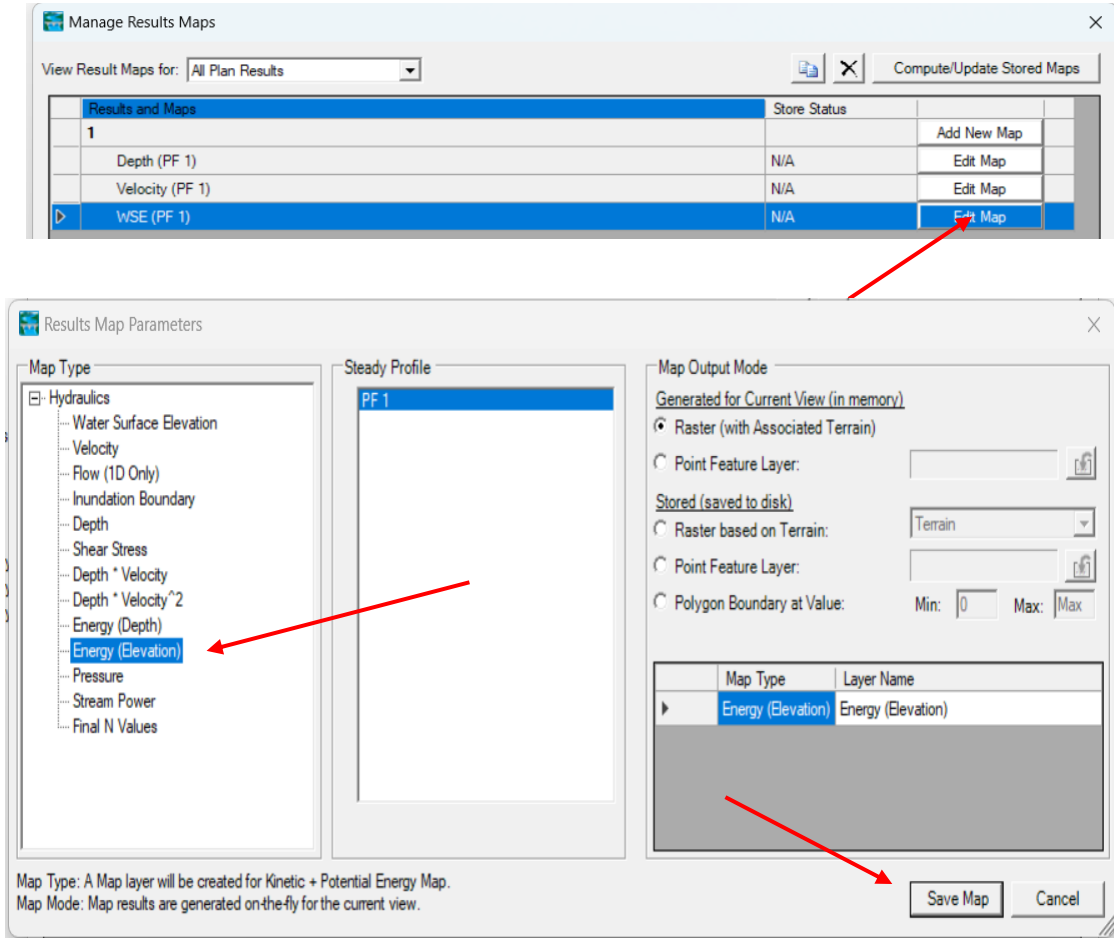
- Open Geometric Data
- Select **River Reach** and Draw the river on the GIS Map created
- Define River and Its reach

View Results in RAS Mapper



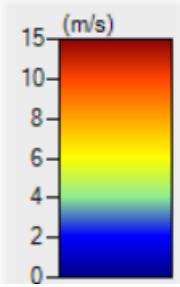
- If you use RAS-Mapper (like in your Assignment)
- You can click on **Depth** and it should give you a vertical view of where the water level rises
- You can view the **Velocity** of the channel as well

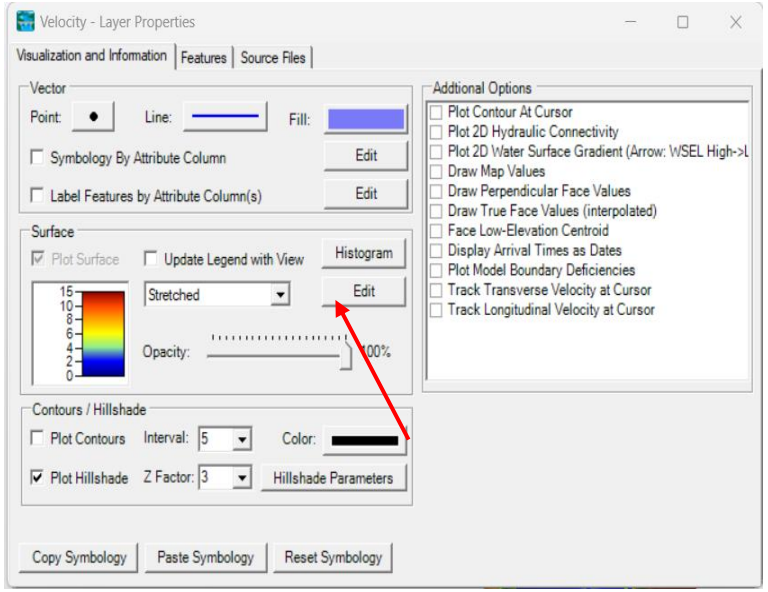
View Results in RAS Mapper

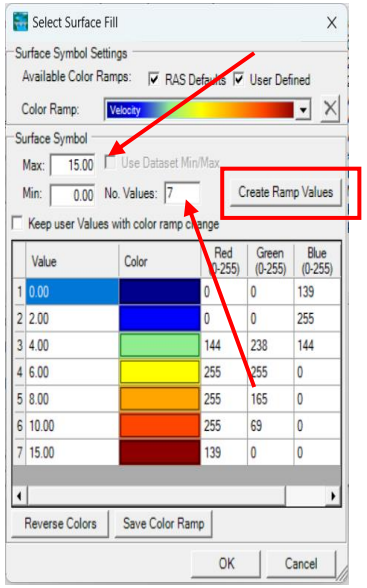


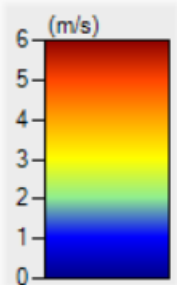
- To display more results maps, right click on **Results** and click on **Manage Results Maps**
- Click **Edit Map** from WSE
- Select the Desired Parameter (in this case **Energy (Elevation))**
- Now click on **Save Map**

View Results in RAS Mapper









- One can change the scale of the results features
- For Example, to change the velocity scale, right click on **Velocity** then click on **Layer Properties**
- In the **Surface** Box, click on **Edit**
- One can now play around and change parameters such as the **Maximum Velocity** and **Number of values** and once satisfied click on **Create Ramp Values**

Thank You